Research Proposal: Evidence of an Emergent "Semantic Self" in Continual Multilingual LLMs

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1. Core Objective

To extend the "Emergent Self" framework from embodied robotics (Jhunjhunwala et al., 2026) to Large Language Models (LLMs). We hypothesize that continual multilingual training forces an LLM to spontaneously partition its neural weights into two distinct groups: an invariant **"Semantic Self"** (language-agnostic reasoning, logic, world knowledge) and a highly plastic **"Linguistic Task"** subnetwork (language-specific syntax, grammar, vocabulary).

2. Literature Grounding & Novelty

Recent NLP mechanistic interpretability research supports our hypotheses, identifying static "Language-Specific Neurons" (Tang et al., 2024) and middle-layer "language-agnostic spaces" (Liu & Niehues, 2025). However, no prior work tracks the *dynamical emergence* of these structures during continual learning. This project will be the first to bridge Embodied AI and NLP by topologically mapping how LLMs partition "thoughts" from "vocabulary" over time, applying the same mathematical principles used to map a robot's physical body schema.

3. Core Hypotheses

- **H1 (The Emergent Self):** Sequential multilingual training induces a persistent subnetwork across language shifts (the "Self"), while task-specific linguistic subnetworks rapidly reorganize.
- **H2 (The Typological Similarity Principle):** The degree of reorganization in the "Task" subnetwork correlates mathematically with structural language distance (measurable via `lang2vec`). Transitions between distant languages (e.g., Hindi → Yoruba → Chinese) will cause massive subnetwork reorganization. Transitions between structurally similar languages (e.g., Chinese → Vietnamese) will result in significantly less neural reorganization.
- **H3 (Directional Layer Localization):** Translation direction dictates *where* the Self topologically anchors in the network.
  - *English → Target Language:* The input is constant, so the Semantic Self will anchor in **early** encoder/processing layers (mirroring the robotics paper's Layer 1 finding).
  - *Target Language → English:* The output is constant, so the Semantic Self will migrate to **later** generation layers.

4. Execution Plan (Adapting the `EmergentRobotSelf` Repo)

We will directly repurpose the lab's existing analytical codebase with minimal modifications.

- **Phase 1: Setup & Data Prep**
  - **Model Selection:** Use a lightweight base LLM (e.g., `SmolLM-135M` or `Qwen2.5-0.5B`) to ensure the $\mathcal{O}(N^3)$ Hungarian matching algorithm remains computationally tractable on dense Transformer layer dimensions.
  - **Shared Reference States:** Extract a parallel corpus of exactly 1,000 sentences (e.g., from FLORES-200) representing the exact same conceptual meaning across English, Hindi, Yoruba, Chinese, and Vietnamese.
- **Phase 2: Continual Training Curriculum**

  - Continually fine-tune the model to a plateau sequentially across our target languages. (Run A: English → Hindi → Yoruba → Chinese. Run B: Hindi → English, etc.)
  - Save model checkpoints at the plateau of every language phase.
- **Phase 3: Subnetwork Extraction (The Code Hook-in)**
  - **Activation Capture:** Pass the 1,000 parallel sentences through the saved checkpoints. Use PyTorch forward hooks to capture the activation traces of the MLP neurons.
  - **Reusing the Repo:** Feed these activations directly into the `EmergentRobotSelf` codebase to compute **cosine similarity**, run **block diagonalization** ($\tau \approx 0.70$), and use **Hungarian matching** to map the separation statistic (Δ) and Persistence Scores.

5. Interventions & Functional Validation (The Vietnamese Test)

To prove the functional importance of these subnetworks (and that they aren't just statistical artifacts), we will use a final Phase 4 (Vietnamese) as our testbed.

- **Similarity Benchmark:** Calculate the Δ (separation statistic) between Chinese and Vietnamese. Compare it to the Yoruba → Chinese transition to mathematically validate H2.
- **Freeze Test:** Identify the "Semantic Self" neurons. Freeze their weights and train on Vietnamese. *Hypothesis: Freezing the Self accelerates learning and acts as a natural, unsupervised defense against catastrophic forgetting of core logic compared to freezing an equal-sized Task subnetwork.*
- **Lesion Test:** Reinitialize the "Semantic Self" neurons. *Hypothesis: Lesioning the Self destroys logical reasoning capabilities (evaluable via translated GSM8K), while lesioning the Task subnetwork merely degrades grammatical fluency.*